Task 2 – Datasheet Literacy

Datasheet-Based Design Analysis: L298N Motor Driver

1. Datasheet Parameters and Their Influence on Circuit Design

The L298N is a dual full-bridge motor driver used to control the direction and speed of two DC motors.(in my case and posibly 4 also)

a) Motor supply voltage

4.5–13.5 V.. this is the voltage suplied to the motors and this helped me choose the battrey for it and also the motors too...

b) Output current rating

1.2 A continuous per channel, 3.2 A peak. And also the peak current of a n20 motor is 0.8 A which is very suitable .. to use the above driver ..

c) Logic input voltage

2.7–5.5 V, with a separate motor supply, VCC covers both 3.3 V and 5 V logic, so I wired it directly to my microcontroller's logic rail with no level shifter. VM and VCC are separate pins on the module, so the motors run from the battery while the logic stays on a clean regulated rail. The motor supply never feeds the microcontroller pins (main advantage ..

2. Condition That Could Damage the Component

Exceeding the 15 V absolute maximum on VM. Above this, the output MOSFETs and internal circuitry can break down, and the damage is permanent. In practice the risk comes from a wrong battery (for example 4S Li-ion at 16.8 V), voltage spikes from the motor, or reversed polarity. The breakout module has no reverse-polarity protection, so reversed battery leads would also destroy it.

To reduce this risk, the motors' current requirements should be checked before use, including their stall current. The driver should be operated within its electrical and thermal ratings. The motor supply voltage must also remain within the permitted range.

How my design avoid the problems are that i will first use a 3A fuse .. and also im using a 7.4 v Lipo battery and buck converter to make sure no wrong volatge go into the motor driver ... and also one very very main advantage is using a buck converter makes sure we get a steady supply og 6v to the driver .. so steady voltage to the motors so having more control on the car caz we are using PID